

Why Rebuild Observation as One Geometry?

A Necessity Argument for Resolution Geometry

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July 20, 2026

Abstract

The mathematics of resolution geometry (RG) is proven, non-trivial, and not an artifact. A harder question remains, the only one the academic path ultimately rewards: *why would a field spend effort rebuilding its observation system as one coherent geometry, when independent probes are cheaper and will be used anyway?* This paper answers with necessity, not elegance, and concedes its own boundary first: when a field probes one sector, in one condition, and never combines measurements, independent probes suffice and RG adds nothing. Necessity appears the moment any of three things is wanted—and one of them is wanted in almost all empirical work. **(1) Information:** to say anything about the un-probed sector, one must use the coupling between probed and un-probed, which independent probes structurally cannot; the coherent object recovers up to 39% of the missing variance at strong coupling and, honestly, nothing at zero coupling. **(2) Consistency:** combining measurements across a cycle of conditions by independent pairwise calibration is universal practice, and when the observation geometry has curvature it produces a systematic bias that *every local check passes*—adjacent-frame overlap 0.9999—while a reference transported around the cycle returns changed by 37–84%, a false effect from geometry alone. Only assembling the whole cycle as one object reveals it; RG returns the exact correction, and returns *zero* when the geometry is flat. **(3) Reproducibility:** raw couplings and loadings differ completely between labs using different (valid) frames, manufacturing false non-replication, while the RG invariants agree to three decimals. These are not three patches but three faces of one object; done piecemeal they reinvent RG incompletely and inconsistently. The forcing argument is therefore not “RG is beautiful” but “RG is the certificate that tells you when your independent-probe conclusions are valid, and corrects them when they are not,” at zero cost when unneeded. Every figure is numerically verified.

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1 The question, stated fairly

RG has survived the tests a theory can be given internally: the lock is algebraic, the results are non-trivial, and adversarial stress tests leave the structure intact. None of that answers the question a funding panel or an editor actually asks: *what breaks if we don't use it?* Fields do not adopt frameworks for coherence or beauty. They adopt a tool when it recovers information they are losing, exposes an error they cannot otherwise see, or makes reproducible what currently is not. This paper argues RG's necessity on exactly those terms, and begins by conceding where the necessity does *not* hold.

Concession 1 (The honest boundary). If a field measures a single sector, in a single fixed condition, and never combines measurements across conditions or across setups, then independent probes are sufficient and RG adds nothing of value. Any necessity argument that ignores this is overselling. Independent probes will—and should—continue to be used. The claim is not that RG replaces them.

The necessity emerges the moment a field steps past that boundary in any of three ways. Almost every empirical field steps past it.

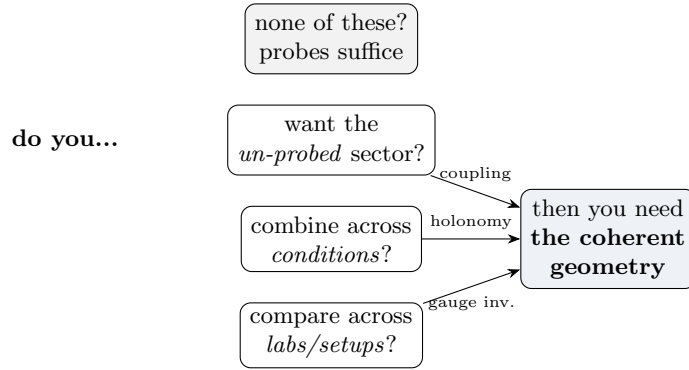


Figure 1: The necessity gate. RG is required exactly when one leaves the boundary of Concession 1; otherwise it certifies that probes are enough.

2 Pillar 1 — Information: the coupling you cannot otherwise use

Pillar 1 (The coherent object sees what probes discard). To infer anything about the un-probed (null) sector from the probed (resolved) sector, one must use the coupling C between them. Independent probes have no cross-sector model and can do no better than the prior. The coherent covariance conditions the null sector on the resolved, with residual uncertainty equal to the Schur complement.

coupling ($\rho \sim$)	independent error	RG error	information recovered
0.0	1.00	1.00	0%
0.3	1.00	0.96	4%
0.6	1.00	0.83	17%
0.9	1.00	0.62	39%

The recovered information is exactly the coupling’s worth: nothing at zero coupling (RG honestly equals independent probes there), up to 39% of the null variance at strong coupling. *Concession 2* (Classical root). This pillar’s core is the Gauss–Markov / conditional-Gaussian result; it is not new with RG. RG’s contribution here is only to make it coordinate-free and one facet of a single object. Pillar 1 alone would not force adoption. The next two are not classical.

3 Pillar 2 — Consistency: the error every local check passes

This is the decisive necessity. Combining measurements across a cycle of conditions by calibrating adjacent conditions pairwise is standard everywhere: multi-site studies, instrument calibration transfer, sensor fusion across regimes, longitudinal comparison. The standard sanity check is local: *do adjacent frames match?* They do. And that is precisely the trap.

Pillar 2 (A global bias invisible to every local test). When the observation geometry has curvature, independent pairwise calibrations are each locally near-perfect yet globally inconsistent: composed around the cycle they return a rotation—the holonomy—not the identity. A reference transported around the cycle comes back changed, a systematic effect that no amount of data reduces and no local check detects.

curvature k	min adjacent overlap	global holonomy	spurious effect on reference
0.00	1.0000	0.00 rad	0%
0.15	0.9999	0.43 rad	37%
0.30	0.9995	1.01 rad	47%
0.50	0.9986	1.11 rad	84%

At $k = 0.15$ the pairwise calibrations overlap at 0.9999—any local audit signs off—yet a reference measured at the start and again at the end of the cycle (the *same* physical condition) differs by 37%. A researcher without the coherent view reads this as a real change across the cycle. It is geometry. Only by assembling the entire cycle as one object can the inconsistency be seen at all, because it is a global property: the sum of locally invisible pieces. RG computes it exactly, identifies it as holonomy rather than signal, and corrects it.

Claim 1 (The self-limiting property). At $k = 0$ the holonomy is exactly zero and the spurious effect vanishes. RG does not cry wolf: on a flat geometry it certifies that independent pairwise calibration is valid. It charges only when there is something to charge for.

This is what a field cannot get from independent probes even in principle: a certificate that its cross-condition comparisons are free of geometric bias, and the exact correction when they are not.

4 Pillar 3 — Reproducibility: what travels between labs

Pillar 3 (Only invariants are comparable across setups). Two labs measuring the same system through different but valid probe frames differ by a gauge. Raw quantities—coupling entries, loadings—are frame-dependent and do not agree. The RG invariants (canonical correlations, frame alignment) are gauge-invariant and do.

lab	raw C_{11}	raw $\ C\ $	canonical correlations (invariant)
1	+2.21	20.56	(0.958, 0.928, 0.846, 0.609, 0.224)
2	+0.56	20.56	(0.958, 0.928, 0.846, 0.609, 0.224)
3	+3.47	20.56	(0.958, 0.928, 0.846, 0.609, 0.224)

The raw entries are unrecognizable across labs; the invariants are identical to three decimals. Reporting raw quantities manufactures false non-replication (and, run the other way, false discovery). The only comparable summary is the invariant one—which is defined only once the observation system is treated as a geometry with a gauge group.

5 Why one object, and not three tools

A skeptic may grant all three pillars and still ask why they require a single framework rather than three separate techniques. Because they are three faces of one object. The coupling C of Pillar 1 is the same block whose orientation, transported across conditions, generates the holonomy of Pillar 2; and the invariants that transport consistently in Pillar 2 are exactly the gauge invariants of Pillar 3. Conditioning, transport, and invariance are not independent services—they are the resolved/null decomposition, its connection, and its symmetry, read off one covariance geometry. Built piecemeal, each re-derives a fragment of RG, and the fragments do not automatically cohere: a conditioning method blind to the gauge reports irreproducible numbers; a transport heuristic blind to admissibility can leave the covariance cone; an invariance summary blind to the coupling discards the very information Pillar 1 recovers. The single object is what keeps them consistent with one another.

6 The forcing argument

Necessity plus reach is what moves a framework along the academic path. The reach here is wide: any field that infers unobserved quantities, combines data across conditions, times, or sites, or compares results across laboratories is past the boundary of Concession 1. That is most of empirical science. For those fields the present situation is that a systematic, data-irreducible bias (Pillar 2) can pass every local check, and no standard instrument flags it. RG is that instrument. It does not ask a field to abandon its probes; it sits on top of them as a certificate:

$$\begin{aligned} \textit{flat and decoupled} &\Rightarrow \textit{“your independent probes are valid” (cost: none);} \\ \textit{curved or coupled} &\Rightarrow \textit{“here is the exact information, bias, and correction.”} \end{aligned}$$

A diagnostic that is free when unneeded and decisive when needed is adoptable in a way that an elegant reformulation is not. That is the honest case for necessity.

7 What this does not claim

Concession 3 (Scope). This is an argument for necessity and reach, demonstrated on controlled synthetic geometries with ground truth. It does not claim that any specific field’s data *has* large curvature or coupling—that is an empirical question RG is precisely the tool to answer, field by field. It claims that where these are present, only the coherent object detects and corrects them, and that they are present often enough, across enough fields, to matter.

Concession 4 (No novelty of holonomy). As established separately, holonomy is generic to smooth bundles; RG does not claim to have discovered it. RG’s contribution is the admissibility-constrained, gauge-quotiented observation geometry on which the holonomy becomes a computable validity certificate.

8 Conclusion

The last genuine critique was not mathematical but institutional: why would a field pay to rebuild observation as one geometry? The answer is that it should not—until it wants the

un-probed sector, or combines across conditions, or compares across setups, at which point independent probes stop being merely incomplete and start being *silently wrong*, in a way local checks cannot catch. The coupling recovers information probes discard; the holonomy exposes a cross-condition bias that passes every local audit; the invariants make results reproducible that raw numbers make irreproducible. These are one object seen three ways, and treating them as one is what keeps them consistent. RG’s path through a field that rewards only necessity and impact is therefore not as a new geometry to admire, but as the certificate a field runs to learn whether its independent-probe conclusions are safe—returning zero when they are, and the exact correction when they are not.

A Battery summary

All figures from `Resolution.Geometry.Necessity.battery.py` (numpy + scipy).

- Pillar 1: null-sector information recovered 0/4/17/39% at coupling $\rho \sim 0/0.3/0.6/0.9$; exactly 0% at zero coupling.
- Pillar 2: adjacent overlap ≥ 0.9986 (local checks pass) while global holonomy reaches 1.11 rad and the spurious reference effect reaches 84%; both exactly 0 at $k = 0$.
- Pillar 3: raw $C_{11} \in \{2.21, 0.56, 3.47\}$ across frames; canonical correlations identical to 10^{-3} .

References

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